

Capacitance Level Transmitter for Liquid Applications

Model: CPLT-1200L



PRODUCT DESCRIPTION:

The CPLT-1200L Capacitance Level Transmitter is designed for continuous level measurement of a wide range of conductive and non-conductive liquids. It works on the principle of detecting changes in capacitance caused by the variation in liquid level. In conductive tanks, the insulated sensing probe acts as one electrode, while the tank wall acts as the other electrode. As the liquid level rises or falls, the dielectric surrounding the probe changes, resulting in a corresponding change in capacitance. The transmitter converts this

capacitance variation into a proportional 4–20 mA output signal for reliable level monitoring and control. For non-conductive liquids or non-conductive tanks, a reference/ground probe is provided to complete the sensing system. The probe is available with PTFE insulation for excellent electrical insulation, chemical resistance and high-temperature applications.

The CPLT-1200L offers convenient field calibration for Empty (4 mA) and Full (20 mA) levels. It is available with rigid probes up to 3 m and flexible probes up to 30 m, with various process connections and mounting arrangements to suit different industrial applications. It is widely suitable for water, edible oils, petrochemicals, acids, chemicals, pharmaceutical liquids and other process media.

SALIENT FEATURES:

- Continuous and accurate level measurement
- Suitable for conductive and non-conductive liquids
- Standard 4–20 mA output
- PTFE-insulated probe for excellent electrical insulation
- Process temperature up to 150°C
- Suitable for corrosive and high-temperature applications
- Easy field calibration for 4 mA and 20 mA
- Rigid probe available up to 2 m
- Flexible probe available up to 30 m
- Reference/ground probe option for non-conductive applications
- Grounding tube option for fuel level measurement
- Various process connections and mounting arrangements
- Compact and rugged industrial construction
- No moving parts – low maintenance
- Suitable for PLC, DCS, SCADA and process control systems
- Customised probe length and material options available

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OPERATING PRINCIPLE:

The CPLT-1200L Capacitance Level Transmitter operates on the principle of capacitance variation between the sensing probe and the reference electrode.

The insulated sensing probe and the tank wall form a capacitor in a conductive tank. As the liquid level rises or falls, the dielectric material surrounding the probe changes, causing a corresponding change in capacitance. The transmitter continuously detects this change and converts it into a proportional 4–20 mA output signal corresponding to the liquid level. For non-conductive liquids or non-conductive tanks, a separate reference/ground probe is provided to form the second electrode.

The 4 mA and 20 mA levels can be easily calibrated in the field to represent the required empty and full tank levels.

APPLICATIONS:

Water & Wastewater • Edible Oils & Vegetable Oils • Petroleum & Fuels • Petrochemicals • Acids & Corrosive Chemicals • Solvents & Process Chemicals • Pharmaceutical Liquids • Resins & Liquid Chemicals • Paints & Coatings • Alcohol & Chemical Solvents • Food & Beverage Processing • Storage & Process Tanks • Conductive & Non-Conductive Liquids • Fuel Level Measurement • Process Monitoring • Automatic Level Control.

PROBE CONSTRUCTION:

The CPLT-1200L is available with different probe configurations to suit various liquid-level measurement applications. The sensing probe is provided with PTFE insulation for reliable electrical isolation, chemical resistance and operation at temperatures up to 150°C.

- Rigid Probe: Suitable for tank depths up to 2 m
- Flexible Probe: Suitable for tank depths up to 30 m
- PTFE Insulated Probe: For conductive and non-conductive liquids
- Reference/Ground Probe: For non-conductive liquids or non-conductive tanks
- Grounding Tube: Available for fuel-level measurement applications
- Custom Probe Length: Available as per tank height and application requirement
- Process Connection: Various threaded and flanged mounting options available
- Probe Material: Selection available as per process media and application requirements

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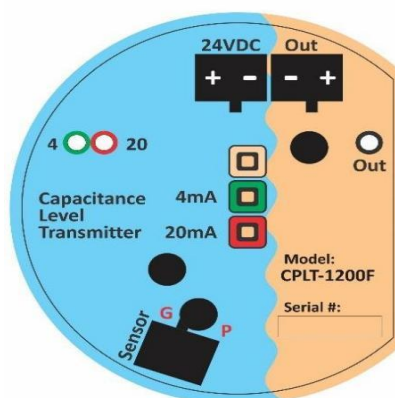
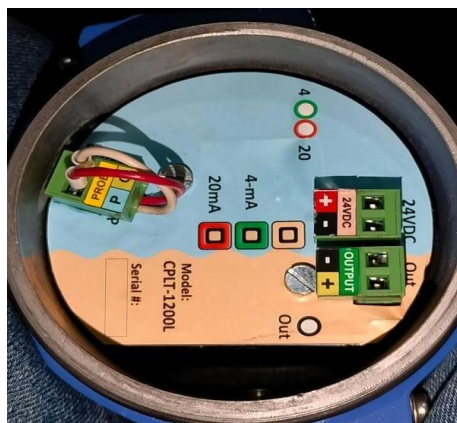
Model: CPLT-1200L



TECHNICAL SPECIFICATIONS: Model: CPLT-1200L

Parameter	Specification
Type	Capacitance Level Transmitter
Supply	24 V DC, 4-Wire
Output	4–20 mA
Permissible Load	500 Ω
Measurement Range – Rigid Probe	200 mm to 2000 mm
Measurement Range – Flexible Probe	Up to 30,000 mm, depending on application
Maximum Pressure	Up to 5 kg/cm ²
Process Temperature	Up to 150°C
Enclosure	Die-Cast Aluminium, Weatherproof / Flameproof
Ingress Protection	IP66
Resolution	0.5 mm
Accuracy	±1% of Full Scale
Process Connection	1" / 1½" BSP / NPT / Flange – Specify
Cable Entry	½" / ¾" ET / NPT – Specify
Probe Insulation	PTFE
Probe Type	Rigid / Flexible
Calibration	Field Adjustable – 4 mA / 20 mA
Reference / Ground Probe	Available for Non-Conductive Applications
Grounding Tube	Optional for Fuel Level Measurement

ELECTRICAL CONNECTIONS: Supply: 24 VDC Output: 4 To 20 mA



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SENSING INSTRUMENTS



Optional Process Indicator Display

Power supply: 230 V AC 50 Hz / 24 V DC @ 3 VA Max Specify.

Out Put: standard SPDT Relay x2 Nos, 4-20 mA Retransmission

For Sensor Supply: 24 V DC 2 250 mA

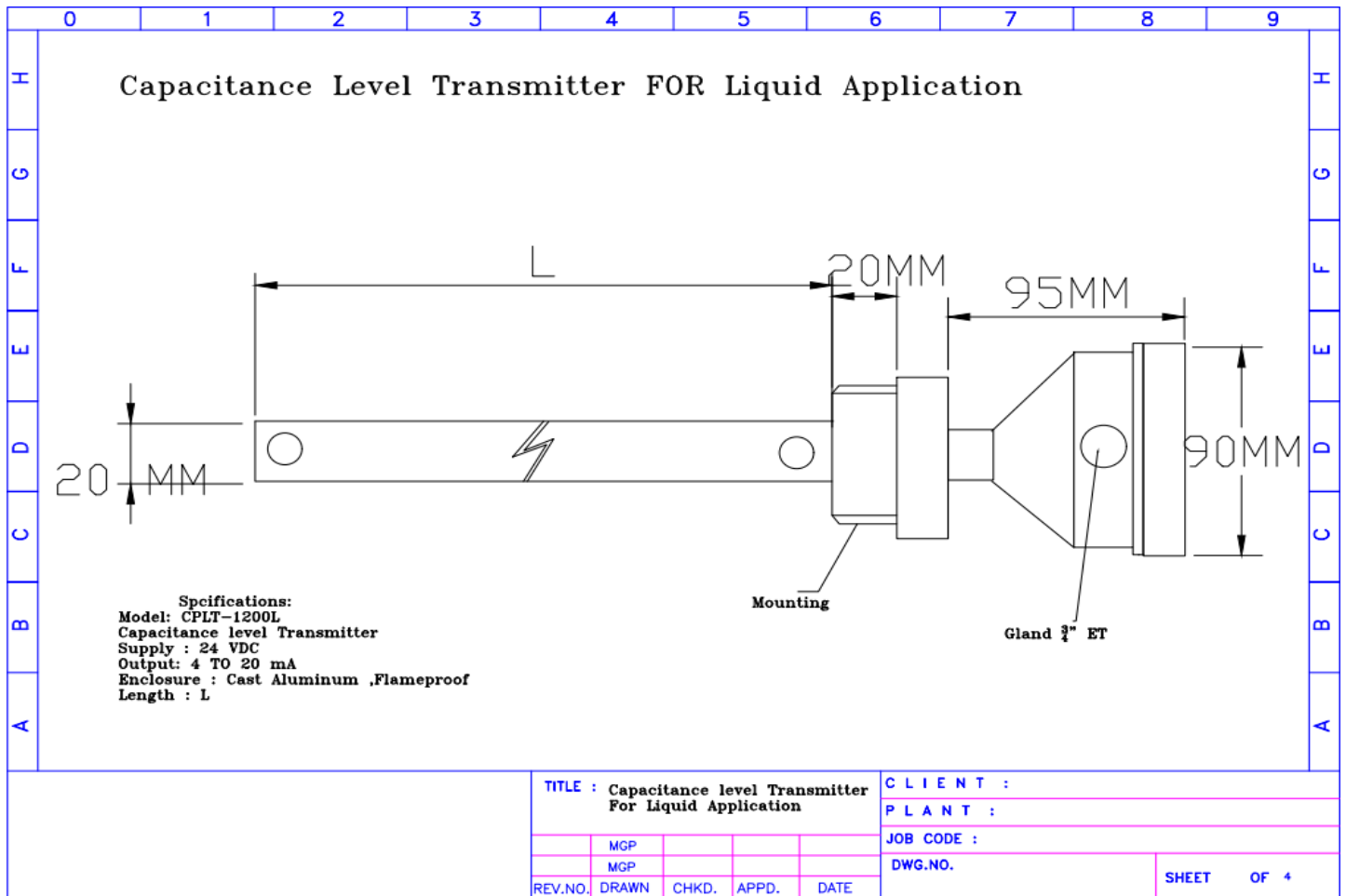
Max. Modbus.

Communication: Programable RS 485 Relay

output: 2 x Potential free C/O contacts



Drawing:



LEVTRON
SENSING INSTRUMENTS

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